

Assignment 6

Stats M148

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Task 1 - time series analysis

Tell me about the model you are using for your data set, why did you chose it? What coefficients are you including and why? Plot or tabulate the coefficients and interpret them. Create a forecast for 2024 and submit a plot of the predictions. The predictive accuracy of your forecast will not be graded (yet). Are there any predictor variables you could use?

We chose the Prophet model for our time series analysis because of the interpretability of its components (trend, seasonality, holidays) and its ability to handle missing data and outliers.

For the monthly model, we just included the trend $g(t)$ and yearly seasonality $s(t)$ components as monthly data does not have the granularity for holidays or weekly seasonality. The components are plotted below:

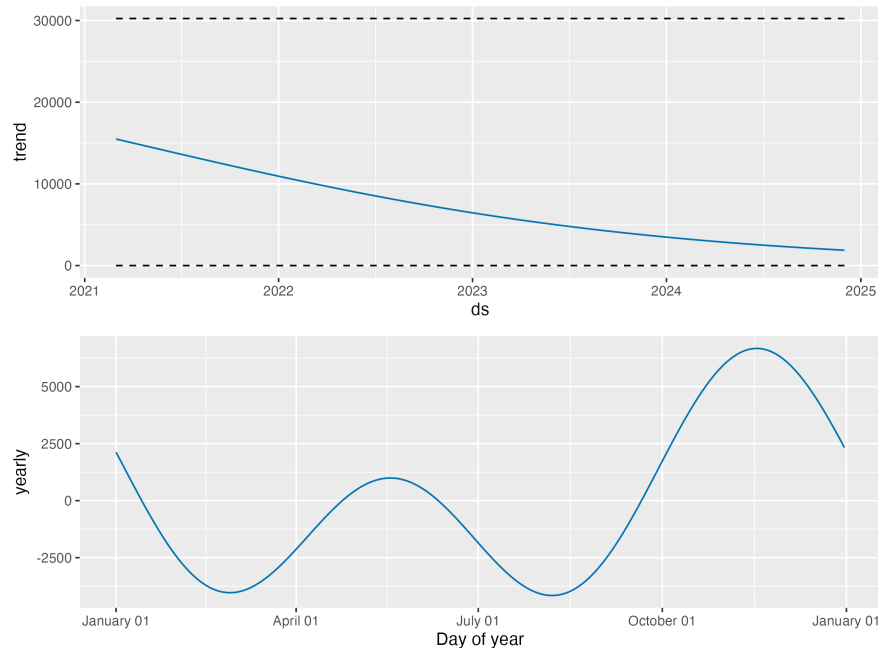


Figure 1: Monthly model components

And here are the predictions for 2024:

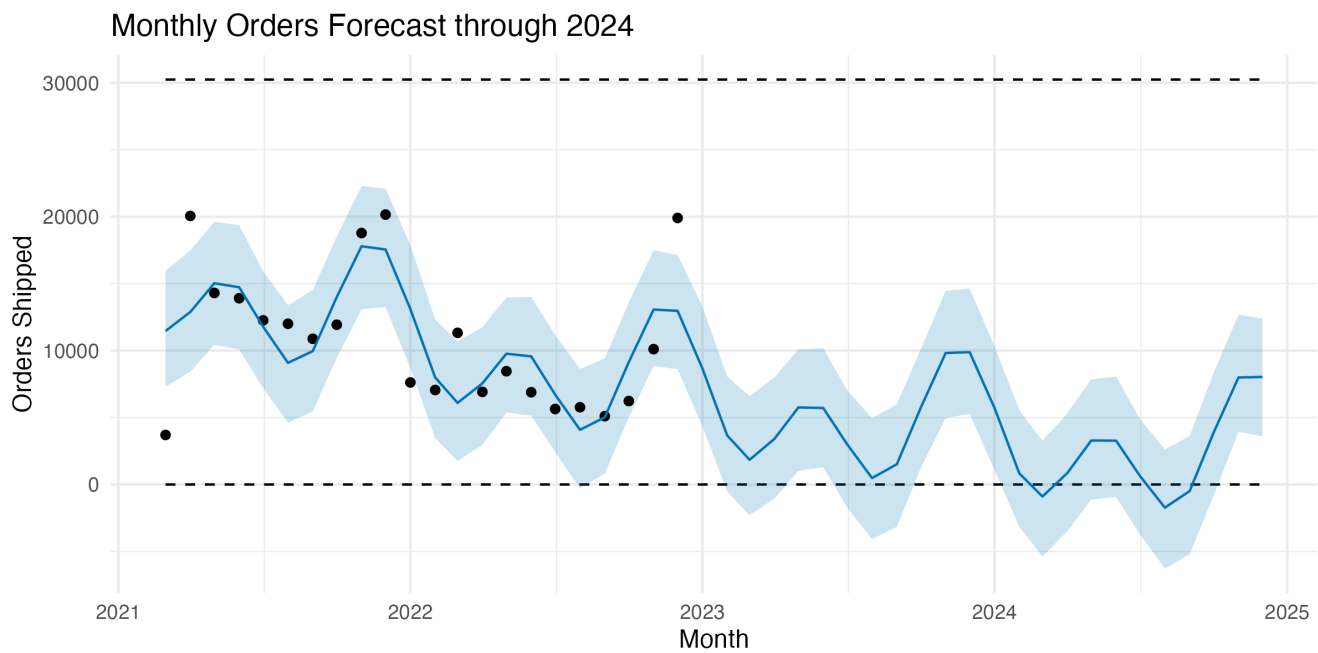


Figure 2: Monthly model predictions

For the daily model, we included the trend $g(t)$, yearly seasonality $s(t)$, weekly seasonality $w(t)$, and US holidays as components. The components are plotted below:

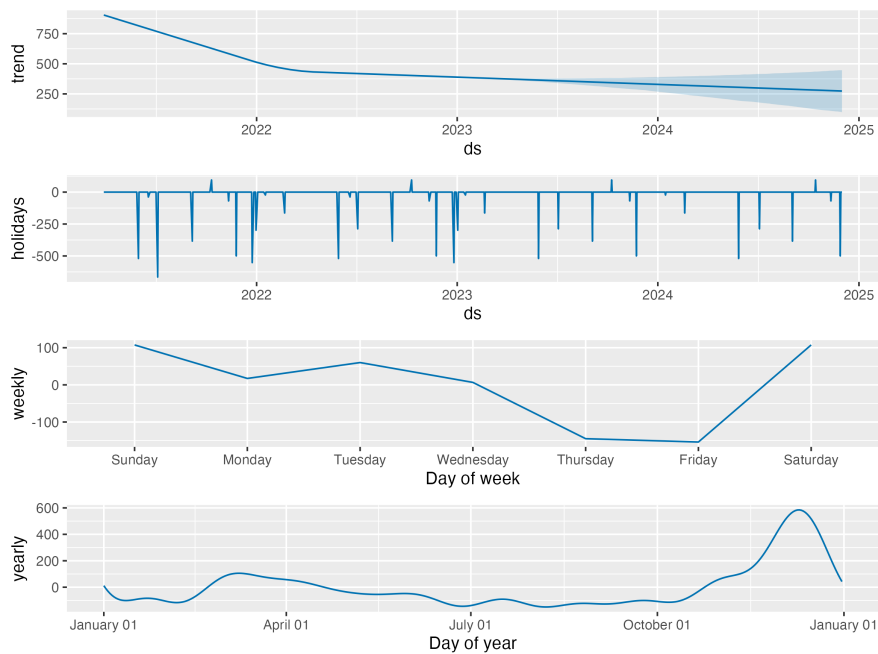


Figure 3: Daily model components

And here are the predictions for 2024:

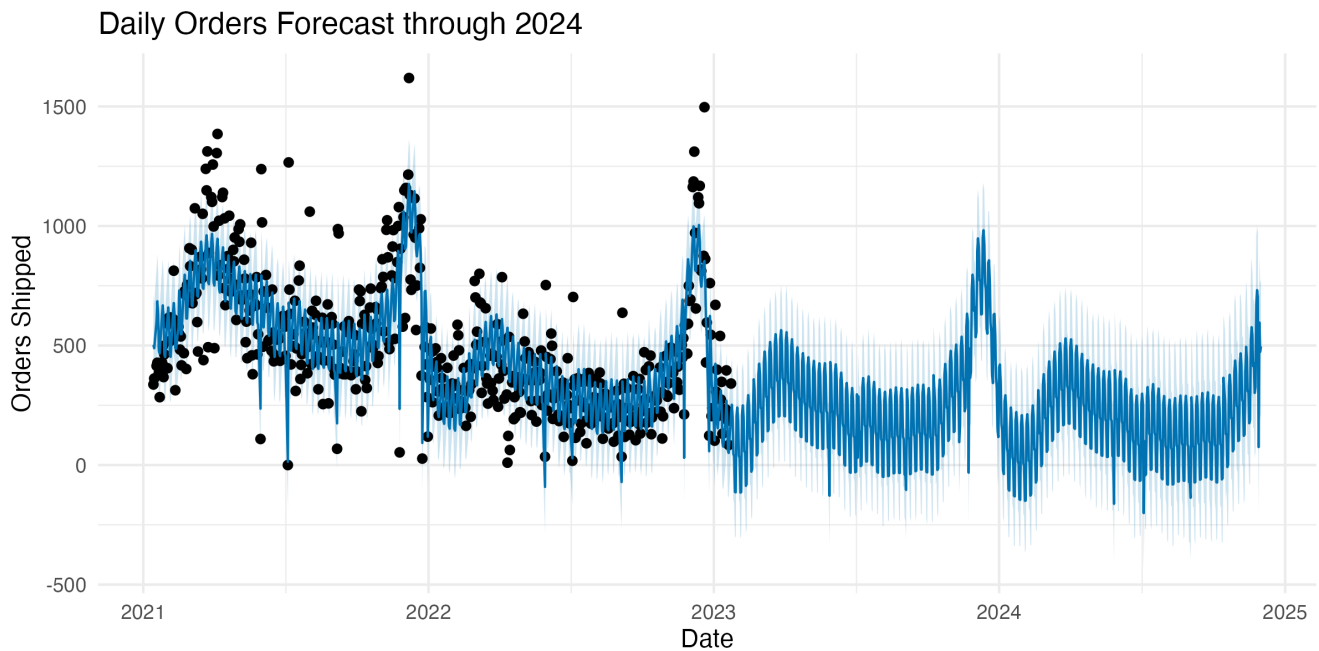


Figure 4: Daily model predictions

Task 2 - predictive modeling reflection

With our late submission, we were able to do pretty well (2nd place) with the first test set competition, albeit with very little tweaking or optimization. One thing to note is that our best predictions came from an XGBoost model trained on a random 10% sample of the data (vs. a model trained on full train set). Hyperparameter tuning did very little to improve performance. A couple ideas I have to improve are as follows:

- Better-informed feature selection (e.g. using LASSO regression or RF feature importance)
- Comparing simpler baseline models (LASSO, logistic regression) to more complex ones (random forest, XGBoost) to see if the added complexity is justified
- More extensive feature engineering: more info on time btw. actions (currently only max, median and mean), indicators for specific action sequences, getting external information on promotions
- Fitting a neural network (RNN, LSTM, or transformer) to capture signal in the sequential nature of the data

Task 3 - dig deeper cont.

We could fit an RNN, LSTM, GRU, or transformer by treating each customer journey as an ordered sequence of actions. Each action would be one step in the sequence and could include the event type,

funnel stage, time since the previous action, and cumulative counts of important actions up to that point.

The main issue is that journeys have different lengths. Some customers may only have a few events, while others may have hundreds. To handle this, we could choose a maximum sequence length, truncate longer journeys to the most recent actions, and pad shorter journeys. We would also use masking so the model ignores padded values.

We would also include time between actions directly, since the same sequence may mean different things depending on whether it happened over minutes or months. A transformed feature like $\log_{1p}(\text{time_since_previous_action})$ would help account for skewed time gaps.-